Chapter 2

Contents

2. History of Microprocessors & Microcontrollers

Microprocessors evolved versions from earlier devices such as vacuum tubes of 1945’s ENIAC, transistors of 1950’s and integrated circuits or ICs from 1960’s.

A microprocessor is a multipurpose programmable device that is driven by a clock. It accepts binary instructions or data from storage memory like RAM/Flash ROM and performs operations on binary data. It consists of registers for temporary storage of binary data or instructions, ALU – arithmetic and logic unit for performing general purpose operations, timing and control unit that utilizes the clock to sequence a set of operational steps required for the execution of instruction that results in an output.

2.1 History of Microprocessors

Here a history two popular processor series is presented, one from Intel and the other from ARM.

2.1.1 Intel Processors

The microprocessors have made their way into different applications and hence processors architectures have evolved over time and specialised processors such as desktop processors, mobile phone processors, microcontrollers etc. have been developed. Here a typical history of these variations is presented without distinguishing between the kind of application.

The birth of first microprocessor started in 1970s through Intel’s 4004 in 1971 which is a 4-bit processor, designed for calculators. It is followed up by Intel’s MCS-4 family and later 8-bit processors 8008 in 1972 and 8080 in 1974.

One of the early popular & widely used 8085 microprocessor was introduced in 1976 which is an 8-bit microprocessor with 16-bit address bus having addressable memory capability of 64K bytes. This followed by MCS-48 family, and the popular 8051 microcontroller and MCS-51 family based on Harvard architecture.

The most popular and the base model of all current Intel’s microprocessors is the 8086 that was introduced in 1978 which is a 16-bit microprocessor with 20-bit address bus having addressable memory capability of 1M bytes. This is the start of the MCS-86 family. The 8-bit external bus version 8088 microprocessor has been introduced in 1979.

As the technology progressed improved and advanced variations of 8086 have subsequently been introduced between 1980 and 1990. The 80186 is the first of improved versions of 8086 but had same 16-data and 20-bit address bus with addressable memory of 16M bytes. The 80286 had 16-bit Data bus with 24-bit address bus & addressable memory of 16M bytes. The 80386, 80486 and several other equivalent variations with 32-bit Data bus, 32-bit address bus & addressable memory of 4G bytes have been introduced.

They are followed up by processors such as Pentium, Pentium Pro, Pentium II, III, IV between 1990 and 2000 with either 32-bit or 64-bit data bus and an increased address bus size of 36-bit with addressable memory capacity of 64G bytes. The P5 microarchitecture based 32-bit processor series were introduced from 1993 with 64-bit data bus and 32-bit address bus with addressable memory capacity of 4GB.

The P6 based Celeron processors have been introduced in 1998 for the low-end/cost consumer pc market. Later other variations in Celeron processors were introduced and has been discontinued from 2023. The P6 based Pentium M series microarchitecture processors had been introduced in 2003 and has been discontinued from 2009.

The atom series processors have been introduced for low-power, low-cost and low-performance applications from 2008 and are still being used. The Itanium processor series was stared in 2001 for high-end server and supercomputer microprocessor applications and has been discontinued in 2020.

The Xeon processor series is another processor series from Intel that was started around 1998 and named from 2001 as Xeon with 32-bit as Pentium II Xeon or Pentium III Xeon and improved to 64-bit in 2004 with different series variations in microarchitectures. The Xeon processors are designed for non-consumer workstation, server and embedded markets and is currently in use.

The first dual core processors are the Pentium D series that have been introduced from 2005 and has been discontinued from 2010. The variations of several dual core Celeron processor variations and atom processor variations from 2008.

The Intel core series was started in 2006 which included intel core with solo – single core, duo – dual core and later expanded to quad – four core processors. The core series includes desktop, mobile and embedded processors. The core 2 processors were the first quad core processor and had been introduced around 2006. The Quad core variants of the atom processors were introduced from 2015.

The current Intel's series of processors the Core i (1^{st} gen) i3, i5 and i7 processors have been introduced in 2010, and a new version is released almost every year since. Core i9 series has been started in the 7^{th} gen of i series in 2017. The 14^{th} generation of the core i series has been introduced at the end of 2023 and beginning of 2024 and is known as Raptor Lake-HX Refresh architecture.

Intel introduced neuromorphic processor through the Lohil processor in 2018 and AI processors through the Core Ultra series – Meteor Lake Microarchitecture in 2023/2024. The mobile core Ultra (Series 1) has been started at during end of 2023 featuring core/core ultra-9, 7, 5 and 3 series and embedded (series 1) in 2024 featuring core ultra-7, 5 and 3 processors. The desktop (series 2) has been started in 2024-2026 featuring core 7, 5 processors, mobile core Ultra (Series 2) during 2024-2026 featuring core/core ultra-9/x9, 7/x7, 5 and 3 series and embedded (series 2) in 2025 featuring core 7, 5 and 3 processors. The mobile core Ultra (Series 3) has been started at during 2026 featuring core/core ultra-9/x9, 7/x7, 5 and 3 series.

Table 2.1: Desktop Processors

Processor	**Architecture/Year**	**Model**
Core i (14th gen)	Raptor Lake-S Refresh/2023-2024	
Core i9		14900KS/14900K/14900KF/14900/ 14900F/14900T
Core i7		14790F/14700K/14700KF/14700/14700F/ 14700T
Core i5		14600K/14600KF/14600/14600T/14500/ 14500T/14490F/14400/14400F/14400T
Core i3		14100/14100F/14100T
Series 2	Arrow Lake-S/2024-2026	
Core Ultra 9		285K/285/285T
Core Ultra 7		270K Plus/265K/265KF/265/265F/265T
Core Ultra 5		250K Plus/250KF Plus/245K/245KF/245/245T/235/235T/ 225/225F/225T

Table 2.2: Mobile Processors

Processor	**Architecture/Year**	**Model**
Core i (14th gen)	Raptor Lake-HX Refresh/2024	
Core i9		14900HX
Core i7		14700HX/14650HX
Core i5		14500HX/14450HX
Series 1	Raptor Lake-U Refresh/Meteor Lake-H/U 2023-2024	
Core/Core Ultra 9		185H
Core/Core Ultra 7		150U/165H/155H/165U/164U/155U
Core/Core Ultra 5		120U/135H/125H/135U/134U/125U/115U
Core/Core Ultra 3		100U
Series 2	Raptor Lake-H/U Refresh/Lunar Lake/Arrow Lake-H/U/HX/Twin Lake-N/2024 - 2026	
Core/Core Ultra 9		288V/270H/285H/290HX Plus/285HX/275HX
Core/Core Ultra 7		250U/265U/255U/268V/266V/258V/256V/

		250H/240H/265H/255H/270HX Plus/ 265HX/255HX
Core/Core Ultra 5		220U/235U/225U/238V/236V/228V/226V/ 220H/210H/235H/225H/245HX/235HX
Core 3		N355/N350
Series 3	Panther Lake/Wildcat Lake 2026	
Core/Core Ultra 9/X9		388H/386H/378H
Core/Core Ultra 7/X7		368H/366H/365/358H/356H/355/360/350
Core/Core Ultra 5		338H/336H/335/325/332/322/330/320/315
Core 3		304

Table 2.3: Embedded Processors

Processor	Architecture/Year	Model
Core i (14th gen)	Raptor Lake-R/2024	
Core i9		14901E/14901TE
Core i7		14701E/14701TE
Core i5		14501E/14501TE/14401E/14401TE
Series 1	Meteor Lake-PS/2024	
Core Ultra 7		165HL/155HL/165UL/155UL
Core Ultra 5		135HL/125HL/135UL/125UL
Core Ultra 3		105UL
Series 2	Bartlett Lake-S/2025	
Core 7		251E/251TE
Core 5		221E/221TE/211E/211TE
Core 3		201E/201TE

Table 2.4: Processors Suffixes & application

Form/Function Type/Segment	Suffix	Optimized/Designed for
Desktop	K	High performance, unlocked
	F	Requires discrete graphics
	S	Special edition
	T	Power-optimized lifestyle
Mobile (Laptop and 2 in 1)	HX	Highest performance, all SKUs unlocked
	HK	Highest performance, all SKUs unlocked
	H	Highest performance

	P	Performance optimized for thin and light laptops
	U	Power efficient
	Y	Extremely low-power efficient
	G1-G7	Graphics level (processors with newer integrated graphics technology)
Embedded	E	Embedded
	UE	Power efficient
	HE	High performance
	UL	Power efficient, in LGA package
	HL	Highest performance, in LGA package

The following Wikipedia links provides a list of microprocessors from different manufacturers, Intel's Core-i series along with other processors based on application and its different generations.

https://en.wikipedia.org/wiki/List_of_Intel_Core_processors
https://en.wikipedia.org/wiki/List_of_Intel_processors
https://en.wikipedia.org/wiki/List_of_microprocessors
https://www.intel.com/content/www/us/en/processors/processor-numbers.html

An exhaustive list of microprocessor manufacturers is provided in the above link. A comprehensive list of popular manufacturers is given below.

- Intel, AMD, NVidia
- Samsung, Apple, Broadcom, QUALCOMM, Amazon, Tesla
- ARM, ST Microelectronics
- NXP Semiconductors – Phillips, Fairchild
- Texas Instruments, Microchip, AVR, Atmel, PIC, Analog Devices
- Motorola, Freescale Semiconductors, Dallas Semiconductor - Maxim
- Zilog, Hitachi, Fujitsu, Matsushita, Harris (Intersil)
- Siemens AG etc.

The Intel's 4004 was manufactured using a 10 µm technology with a single core. The latest Intel's Core Ultra 9 uses a 3nm technology and has 8 performance cores and 16 efficient cores totalling to 24 cores. The Intel's processors are typically CISC – Complex instruction set computer architectures. The details of Intel® Core™ Ultra 9 Processor 285T can be found at the following link. Interested readers may look at the features and technologies employed in this processor. They are provided in the Appendix.

https://www.intel.com/content/www/us/en/products/sku/241059/intel-core-ultra-9-processor-285t-36m-cache-up-to-5-40-ghz/specifications.html
Summary of Intel Microprocessor & Microcontroller is presented in the below Table 2.5.

Table 2.5: Intel Processors

Series	Processor Variations
Early Series	4004, MCS-4 family, 8008, 8080, 8085, MCS-49 family, 8051, MCS-51 family, MCS-96 family, 3000 families, 8086, 8088, 80186, 80286, 80386, 80486
Pentium/P5/P6 Series	P5, Pentium, P6 - Pentium M, Pentium D, Pentium II, Celeron, Celeron M, Pentium III, Pentium II/III Xeon, Pentium 4, Itanium (64-bit)
Intel Core Series	Intel Core, Core 2
Atom Processors	Silverthorne, Lincroft, Diamondville, Pineview, Cedarview, Centerton, Brianwood, Avoton
Intel Core i Series	Core i3 (1^{st} gen – 14^{th} gen), Core i5 (1^{st} gen – 14^{th} gen), Core i7 (1^{st} gen – 14^{th} gen), Core i9 (7^{th} gen – 14^{th} gen)
Xeon Series	Pentium II Xeon, Pentium III Xeon, Xeon UP/DP/MP, 3000, 5000, 7000, Xeon E3, E5, E7, Xeon D, Xeon W, Xeon Bronze, Xeon Silver, Xeon Gold, Xeon Platinum, Sierra Forest-SP, Grand Rapids – AP/SP/D/WS
Core Ultra Series	Three Series of Core Ultra 3, Core Ultra 5, Core Ultra 7, Core Ultra 9 – Series1, Series 2, Series 3.

2.1.2 ARM Processors

ARM is another popular process which is widely used by many manufacturers as it is an IP based company that licenses instruction set architecture designs from 1981 and is not a microprocessor or a microcontroller chip manufacturer. The ARM processors are typically RISC– Reduced instruction set computer architectures. ARM processors are being used in many applications – consumer marker - laptops, desktops, mobiles, embedded applications and super computers similar to Intel. The ARM IP cores are used by many manufacturers mentioned in the above comprehensive list of popular manufacturers presented in 2.1.1 and hence have a wider scope and reach.

ARM was initially known as Acorn computers whose first design was BBC Micro introduced in 1981. The design series from ARM started through ARM1 in 1985 with ARMv1 architecture, ARM2, ARM3 etc. in 1987 with ARMv2 architecture. It is followed by ARMv3 architecture with ARM6 and the ARM7. Later it is followed by ARM8 with ARMv4 architecture & most popular classic ARM processor ARM7 TDMI and ARM9 TDMI with ARMv4T architecture having Thumb instructions in addition to ARM instructions. Later several other classic architectures were introduced such as ARMv5TE with variations of ARM7 and ARM9 and ARM10. The last of the classic processor architecture introduced was ARMv6 through the ARM11.

The current prevalent cortex series processors are enhanced versions of ARMv6. These processors are Cortex A series for the applications like mobile etc., Cortex R series for the real time applications and Cortex M series processors for microcontroller applications.

Several of the Cortex M series processors for microcontroller applications, like the Cortex M0, M0+ M1 have used architecture version ARMv6-M, Cortex M3 using ARMv7-M, and Cortex M4 and M7 using ARMv7E-M. The Cortex M23, M33 used the architecture versions of ARMv8-M and Cortex M55 and M85 using ARMv8.1-M architecture. All these Cortex M processors were 32-bit processors.

Several of the Cortex R series processors for real time applications, like the Cortex R4, R5, R7, R8 through architecture versions ARMv7-R, the Cortex R52, R82 through ARMv8-R. All the Cortex R processors were 32-bit except for the R82 which was the first 64-bit real time processor.

Several of the Cortex A series processors for mobile applications like the Cortex A5, A7, A8, A9, A12, A15, A17 have used architecture versions ARMv7-A and used 32-bit structure. The Cortex A32 was introduced with ARMv8-A using a 32-bit structure. Later using the same ARMv8-A architecture A35, A53, A57, A72, A73 were introduced with 64/32-bit structure and A34 with a 64-bit structure. Later Cortex A55, A75, A76, A77, A78, were introduced in ARMv8.2-A architecture with 64/32-bit structure and A65 with 64-bit structure. Cortex-A510, A710, A715 were introduced with ARMv9-A architecture, Cortex A320, A520, A720, A725 with ARMv9.2-A architecture with 64-bit structure.

Arm has also started introducing the custom versions with Cortex-X1 through ARMv8.2-A using 64/32-bit structure for purely performance instead of a balance of power, performance and area. Similarly, X2, X3 were introduced using ARMv9-A architecture, X4 and X925 using ARMv9.2-A architecture and C1-Ultra using ARM9.3-A architecture. Later the Cortex-X930 has also been introduced.

ARM has a SecurCore 32-bit series - SC100, SC000, SC300 being introduced from ARMv4T, v6-M and V7-M respectively that are designed with anti-tampering features.

Arm has then introduced the Neoverse processors N, V & E series.

N1 was introduced through ARMv8.2-A using 64/32-bit structure. N2 was introduced through ARMv9.0-A using 64-bit structure. Similarly, N3 was introduced through ARMv9.2-A using 64-bit structure.

E1 was introduced through ARMv8.2-A using 64-bit structure. E2 was introduced through ARMv9.0-A using 64-bit structure.

V1 was introduced through ARMv8.4-A using 64-bit structure. V2 was introduced through ARMv9.0-A using 64-bit structure. V3 was introduced through ARMv9.2-A using 64-bit structure.

The latest release from Arm has the C1 series with ARMv9.3-A architecture having C1-Ultra, C1-Premium, C1-Pro, C1—Nano variations.

All the above are core IP designs and are used by several manufacturers for developing their custom microprocessors or microcontrollers. Readers are encouraged to see the respective manufacturers products for usage of ARMs core IP designs. References to sample of products developed using Cortex-M3, M4, M7, A72 & A78 are presented below.
https://en.wikipedia.org/wiki/ARM_Cortex-M#Cortex-M3
https://en.wikipedia.org/wiki/ARM_Cortex-M#Cortex-M4
https://en.wikipedia.org/wiki/ARM_Cortex-M#Cortex-M7
https://en.wikipedia.org/wiki/ARM_Cortex-A72
https://en.wikipedia.org/wiki/ARM_Cortex-A78

Table 2.6: ARM Processors

Series	Processor Variations
Classic - Legacy	ARM1, ARM2, ARM3, ARM7 (ARM7-TDMI), ARM8, ARM9, ARM10, ARM11
Cortex-M	M0, M0+, M1, M3, M4, M7, M23, M33, M35, M52, M55, M85
Cortex-R	R4, R5, R7, R8, R52, R52+, R82
Cortex-A	A5, A7, A9, A15, A17, A32, A34, A35, A53, A55, A65, A57, A72, A73, A75, A76, A77, A78, A510, A710, A715, A320, A520, A720, A725
Cortex-X	X1, X2, X3, X4, X925, X930
SecurCore	SC100, SC000, SC300
Neoverse	E1, E2 N1, N2, N3 V1, V2, V3
C1	C1-Ultra, C1-Premium, C1-Pro, C1-Nano

ARM is in the transitioning process to manufacture its own hardware chips in the future. The following Wikipedia links provides information on the ARM processor series.
https://en.wikipedia.org/wiki/ARM_architecture_family
https://en.wikipedia.org/wiki/List_of_ARM_processors
Summary of ARM processors is presented in the table 2.6.

2.2 Introduction to 8085

One of the popular 8-Bit microprocessor from Intel is the 8085. The features of 8085 are as follows.

- 8-Bit Microprocessor/ALU
- Multiplexed Address/Data Bus:
    - 8-Bit Data Bus AD0 - AD7
    - 16-Bit Address Bus A8-A15
      (Lower 8-Bits Shared with Data Bus – AD0-AD7)
- Memory addressing 2^{16} = 65536 Bytes = $2^6 * 2^{10}$ = 64 KB.
- Simple architecture
    - Registers: A, B, C, D, E, H, L; Stack Pointer – SP, Program Counter – PC, Instruction Register – IR, Register pairs - BC, DE, HL
      Flag Register with 5 status flags:
      S – Sign, Z – Zero, AC – Auxiliary Carry, P – Parity, CY - Carry
    - Serial I/O Control (SID, SOD)
    - Interrupt Control – 5 Interrupts (Trap, RST7.5, RST6.5, RST5.5, INTR)
    - Instruction Decoder & Machine Cycle encoding
    - Increment/Decrement Latch
    - Timing & Control Unit
- Adequate Instruction Set - 74 Instructions, 246 Hex Codes.
    - Addressing Modes: Register, Immediate, Direct, Register Indirect & Implicit
    - Instruction size – one byte, two bytes, three bytes
- 3MHz Clock, Variations available.
- +5V DC Power
- 40 Pin DIP

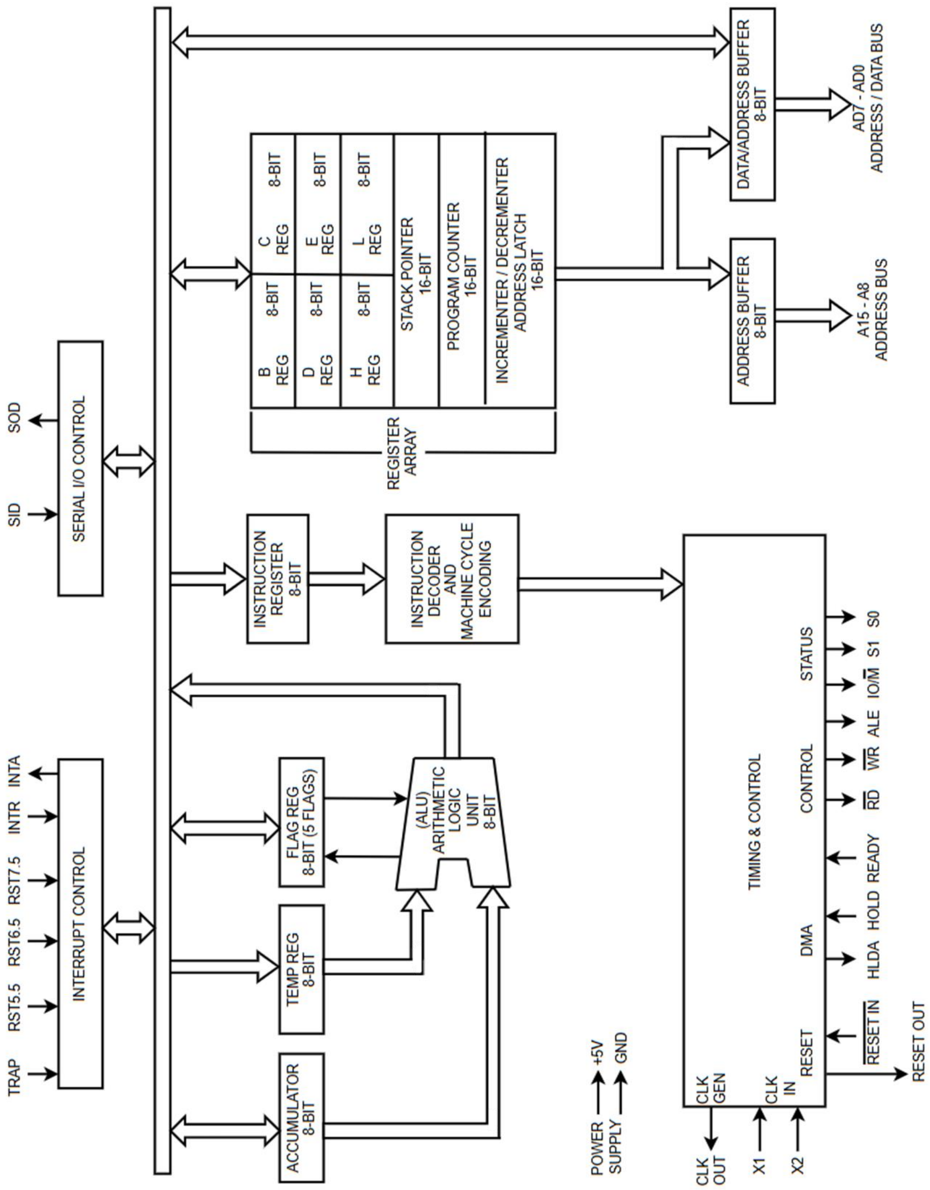

Fig 2.1: Internal Architecture of 8085

Additional information may be found in the following References:

- Microprocessor Architecture, Programming and Applications with the 8085 - Ramesh Gaonkar, 5th Edition, Penram International Publishing Pvt. Ltd. 2007.
- Fundamentals of Microprocessors and Microcomputers – B RAM, 5th Revised Edition, Dhanpat Rai Publications 2001.
- https://en.wikipedia.org/wiki/Intel_8085

2.3 Introduction to 8086

One of the popular 16-Bit microprocessor from Intel is the 8086. The features of 8086 are as follows.

- 16-Bit Microprocessor/ALU
- Multiplexed Address/Data Bus:
  - 16-Bit Data Bus AD0 - AD15
  - 20-Bit Address Bus A16-A19
    (Lower 16-Bits Shared with Data Bus) AD0-AD15
- Memory addressing 2^{20} = 1 MB.
- Improved architecture w.r.t 8085
  - Divided to 2 sections: Bus Interface Unit – BIU, Execution Unit – EU
  - Overlapped fetch and execution using 6-byte queue
  - Memory Segmentation
  - Odd and Even Bank Memory organization for 16-bit access
  - 20-Bit Effective Address computing hardware.
  - Registers:

    General Registers

        AX – AH, AL, - Accumulator
        BX – BH, BL, - Base
        CX – CH, CL, - Counter
        DX – DH, DL - Data

    Flag Register – PSW: Program Status Word

    Flag Register with 6 status flags & 3 control flags:

        Status Flags
        SF – Sign, ZF – Zero, AF – Auxiliary Carry,
        PF – Parity, CF – Carry, OF – Overflow
        Control Flags
        DF – Direction, TF – Trap, IF - Interrupt

    Index Registers

        SI – Source Index
        DI – Destination Index

    Pointer Registers

        SP – Stack Pointer
        BP – Base Pointer
        IP – Instruction Pointer

    Segment Registers

        CS – Code Segment

DS – Data Segment

ES – Extra Segment

SS – Stack Segment

- More powerful Interrupt Control structure
  (NMI, INTR, extendable interrupt vector using 8259)
- Instruction Decoder & Machine Cycle encoding
- Timing & Control Unit
- Higher operating speed
- Improved programming flexibility

- More powerful Instruction Set - 117 including string instructions.
  - Addressing Modes: totalling 24 modes
    Register, Immediate, Direct, Register Indirect & Implicit, Indexed, Register Relative, Based Index, Relative Based Index, Intra segment Direct, Intra segment Indirect, Intersegment Direct, Intersegment Indirect, Intersegment Register Indirect
  - Instruction size – one byte, two bytes, three bytes, four bytes.
- 5 to 10 MHz Clock, Variations available.
- +5V DC Power
- 40 Pin DIP

Additional information may be found in the following References:

- Advanced Microprocessors & Peripherals – A K Ray, K M Bhuruchandi, 2nd Edition, McGraw Hill, 2010.
- Microprocessors and Interfacing, Programming and Hardware – Douglas V Hall, 2nd Edition, McGraw Hill. 2005.

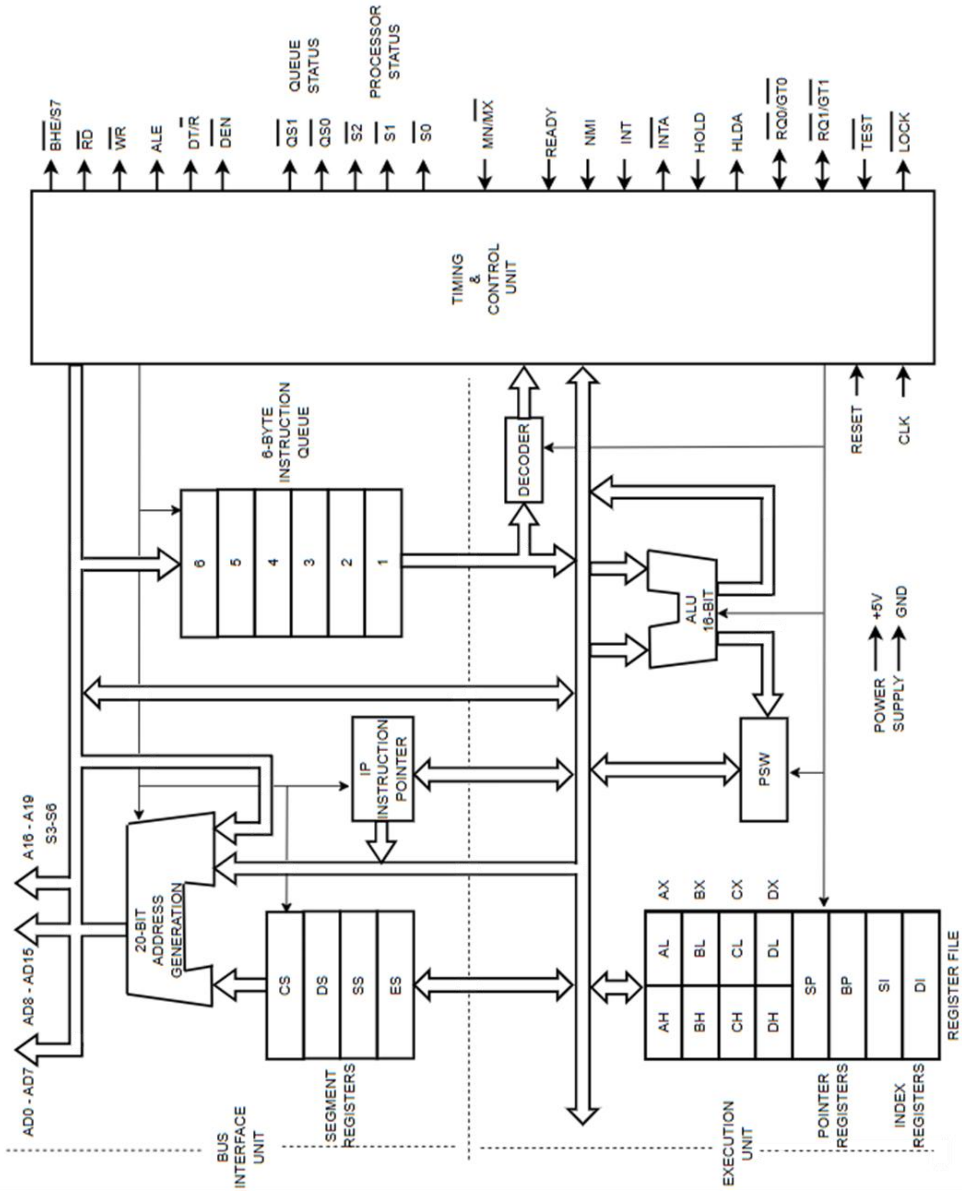

Fig 2.2: Internal Architecture of 8086

2.4 Microprocessors vs Microcontrollers

The microprocessors and microcontrollers typically differ based on their application. Microprocessors as the name suggests is used mainly for data processing applications and generally consists of only processing elements and does not include peripheral components. The Microcontroller on the other hand is used typically for embedded applications requiring controlling features in addition to processing functions and hence includes many built in peripherals and the difference is presented in Table 2.7.

Table 2.7: Microprocessors vs Microcontrollers

Microprocessors	Microcontrollers
Designed for processing purposes	Designed for embedded control purposes
Memory (RAM/ROM) is external to Microprocessor. Must be interfaced.	Microcontrollers have internal memory (RAM/ROM) and can be interfaced with external memory but generally not required as variations exist with higher capacities.
I/O Port pins are not present. Must be interfaced through programmable peripheral devices	I/O Port pins are present. Additional Port pins can be interfaced but generally not required as variations exist with higher I/O ports.
Limited number of external interrupts. Interrupt capability is expanded through external Interrupt controllers or a cascade of controllers.	Reasonable number of external interrupts. Most of the peripheral components are internal and hence generally doesn't require higher external interrupts. Interrupt capability can be expanded through external Interrupt controllers.
Generally, have greater number of internal vectors to accommodate all external and internal, software or hardware interrupts.	Generally, have greater number of internal interrupts and associated vectors with limited software interrupts, Have a limited vectors for external hardware interrupts.
General purpose Timers/counters are not present. Must be interfaced through programmable external timers.	General purpose Timers/counters are present internally. Additional timers can be interfaced but generally not required as variations exist with more timers.
Peripherals such as Capture/Compare/PWM, ADC, DAC are not present and must be interfaced.	Peripherals such as Capture/Compare/PWM, ADC, DAC are present internally.

Communication protocol features such as, Serial Port – USART, RS485, I2C, SPI, CAN, USB, Ethernet are not present and must be interfaced.	Communication protocol features such as, Serial Port – USART, RS485, I2C, SPI, CAN, USB, Ethernet are generally present.
PCB must include all the peripherals, memory, processors and any sensory components making it bulky, more power consuming and slow as everything is interfaced externally to processor	PCB generally include only the microcontroller and sensory components as all peripherals, memory is internal to microcontroller making it compact, less power consuming and comparatively fast as most of the components are internal.

Readers are encouraged to see a microprocessor-based time and temperature system and a microcontroller-based time and temperature system that is presented in the below reference for graphical view of difference between a microprocessor based and microcontroller-based system.

- Microprocessor Architecture, Programming and Applications with the 8085 - Ramesh Gaonkar, 5th Edition, Penram International Publishing Pvt. Ltd. 2007.

2.5 Introduction to 8051

One of the popular 8-Bit microcontroller from Intel is the 8051. The features of 8051 are as follows.

- 8-Bit Microcontroller/ALU
- Multiplexed Address/Data Bus:
    - 8-Bit Data Bus AD0 - AD7
    - 16-Bit Address Bus A8-A15
      (Lower 8-Bits Shared with Data Bus – AD0-AD7)
- Memory addressing 2^{16} = 65536 Bytes = $2^6 * 2^{10}$ = 64 KB.
- 4K internal ROM
- 128 Byte RAM divided to three parts
    - 4 Register Banks, each with 8 Registers. 4 Banks = 32Bytes
    - 16 Bytes of bit addressable memory
    - 80 Bytes of general-purpose memory
- Each register (except PC) has memory location and can be addressed.
- Harvard architecture
    - Registers: A, B;
      4 Register Banks with 8 Registers (R0 - R7) each = 32 Registers
      16-Bit Program Counter – PC, 16-Bit Data Pointer – DPTR
      8-Bit Stack Pointer – SP

    Flag Register with 4 status, 1 Use flag, & 2 Bank select bits – RS1 RS0
        OV – Overflow, AC – Auxiliary Carry, P – Parity, CY - Carry
    - 4 - Four Ports – 8 Bit I/O Ports = 32 programmable I/O Pins.
    - 2 - Two 16-Bit Timers & Counters – T0 & T1
    - Full Duplex Serial Data Receive – RX /Transmit - TX with Serial Buffer Register - SBUF
    - Interrupt Control – 5 Interrupts {2 external (INT0, INT1), 3 internal (Serial, T0, T1) }
    - Instruction Decoder & Machine Cycle encoding
    - Timing & Control Unit
- Adequate Instruction Set - 74 Instructions, 246 Hex Codes.
    - Addressing Modes: Register, Immediate, Direct, Register Indirect & Implicit, Relative & Indexed Addressing.
    - Has internal RAM memory to memory transfer instruction.
      (Not present in 8085 or 8086).
    - Instruction size – one byte, two bytes, three bytes
- 1-20MHz Clock, Variations available.
- +5V DC Power
- 40 Pin DIP

Additional information may be found in the following References:

- The 8051 Microcontroller – Kenneth J. Ayala 3rd Edition, Cengage Learning, 2010.
- The 8051 Microcontroller & Embedded Systems – Mazidi, Mazidi, McKinlay, 2nd Edition, Pearson Education, 2009.

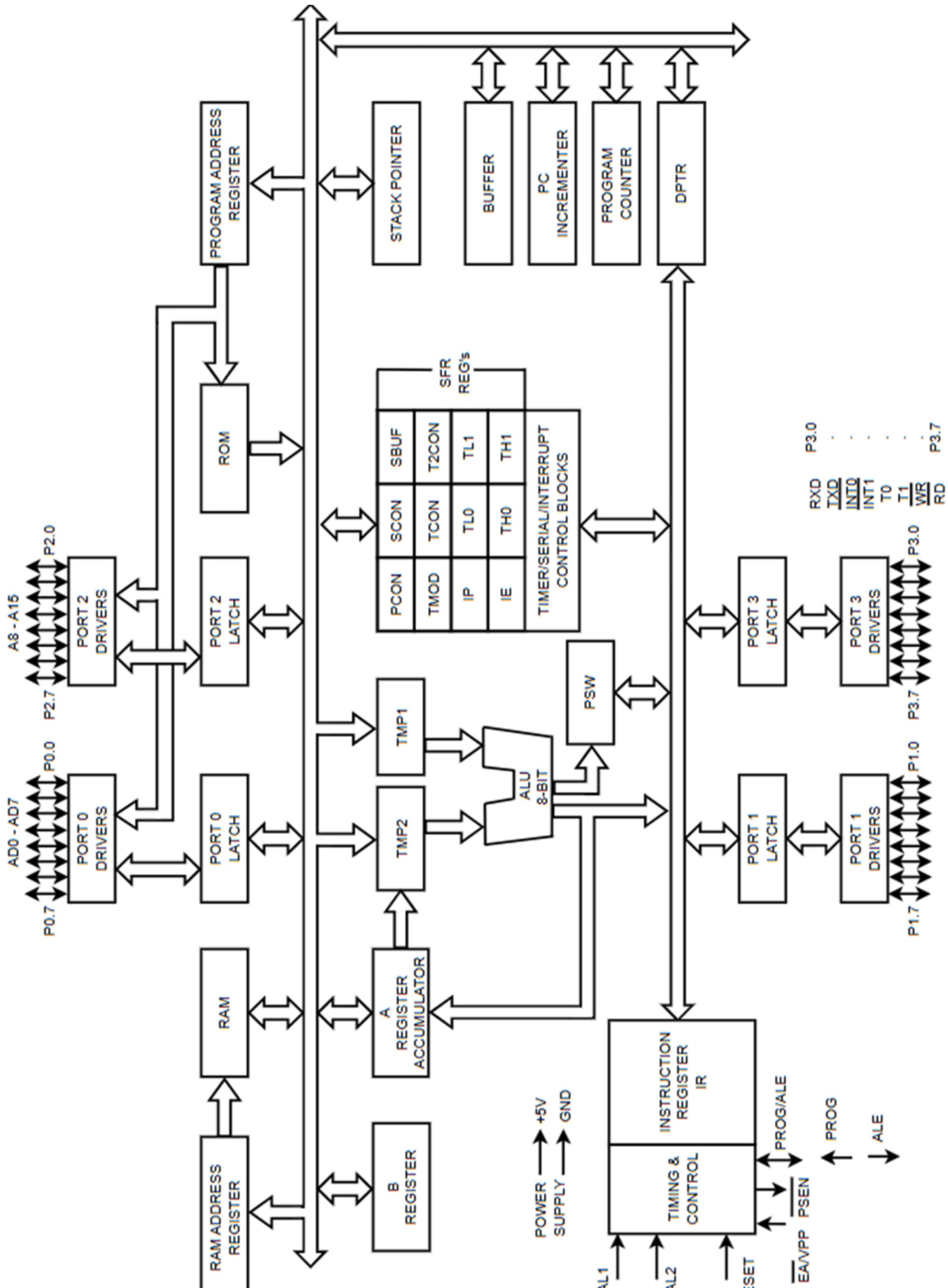

Fig 2.3: Internal Architecture of 8051

2.6 History of Microchips Microcontrollers

2.6.1 PIC 16 Series

The PIC16 series microcontrollers are 8-bit microcontrollers with 14-bit opcode and hence 14-bit wide code memory that are popular for low end applications. Some of the latest PIC16 MCUs are PIC16F131xx, PIC16F132xx, PIC16F152xx, PIC16F171xx, PIC16F175xx, PIC16F180xx, PIC16F181xx etc. The xx number varies based on the memory size and number of peripheral I/O pins.

2.6.2 PIC 18 Series

The traditional PIC 18 series microcontrollers are 8-bit microcontrollers with 16-bit opcode and hence 16-bit wide code memory and have been presented in chapter 3 with introduction to PIC 18 architecture and several of its family variations.

The new variations in PIC18 include the Q series which includes the intelligent and analog features with computation and zero crossing detectors in addition to Core Independent Peripherals – CIPs (ex: Configurable Logic Cells, DMAs, PWMs etc.) that are highly versatile/flexible and independent of CPU core. Variations in Q series include Q10, Q20, Q24, Q35, Q40, Q41, Q43, Q71, Q84 etc.).

2.6.3 PIC 24 Series

The PIC 24 series from Microchip are 16-bit data MCUs with 24-bit address bus capacity. The PIC24 series includes GA, GB, and GU/GL/GP MCUs.

The PIC24FJ GA MCU series are used for Low-Power, Secure and Smart Embedded, extremely cost-effective designs for Internet of Things (IoT) sensor nodes, portable medical devices and industrial control applications and comes with Core Independent Peripherals (CIPs). The PIC24FJ GB MCU series are used to Enable Secure, Low-Power and OTA-Upgradeable High-performance Applications. The PIC24FJ GU/GL/GP MCU series are used for Low-Power MCUs with Segment LCD, USB, Secure IoT, Safety, Sensor Interfacing applications and comes with Core Independent Peripherals (CIPs).

The PIC IOT WA development board - EV54Y39A is designed for use with AWS cloud and PIC IOT WG development board - AC164164 is designed for use with Google cloud. Both these IOT development boards are based on the PIC24FJ128GA705 MCU.

2.6.4 PIC 32 Series

The PIC 32 series from Microchip are based on several CPU architecture variations. PIC 32 variations are available using ARM Cortex-M CPUs, MIPS 32 core architecture and PIC 32A CPUs that have Double-Precision Floating Point Unit (DP-FPU) and High-Speed Analog structure.

The PIC 32's with ARM Cortex-M CPUs is based on Arm Cortex-M0/M0+, M3/M4, M7, M23/M33 CPUS from ARM. PIC 32CK, 32CM, 32CX, 32CZ series. Higher performance devices are based on Cortex-M7, and lower end performance devices are based on M0/M0+ architecture.

Reference

https://www.microchip.com/en-us/products/microcontrollers/32-bit-mcus/pic32-sam

PIC 32A based MCUs are preferred for math intensive applications and include Double-Precision Floating Point Unit (DP-FPU) and High-Speed Analog structure. These are the PIC 32AK series.

https://www.microchip.com/en-us/products/microcontrollers/32-bit-mcus/pic32a

The MIPS32 based MCUs are optimized for real-time operations and ensures quick response times and used for industrial control, automotive systems, consumer electronics and other applications. These are the PIC 32 MK MM MX MZ series.

https://www.microchip.com/en-us/products/microcontrollers/32-bit-mcus/pic32m

2.6.5 SAM Series

The SAM series from Microchip are based on ARM CPUs. The SAM series MCUs are currently based on Arm Cortex-M0/M0+, M3/M4, M7, M23/M33 CPUS from ARM. One of the popular SAM series MCU is the Atmel SAM3X8E that is based on ARM Cortex-M3 CPU. There are several sub series of SAM like SAM 3, SAM 4, SAM C, SAM D, SAM E, SAM G, SAM L, SAM S and SAM V. High-performance series. SAM E7x and SAM E7x MCUs are based on Cortex-M4F CPU and are most widely used in different applications higher performance. The highest performance MCUs are available in SAM V7x, SAM S7x that uses Cortex M7 CPU. The SAM C and SAM D series based on Cortex-M0/M0+ CPUs are used widely used in several of the low-end performance applications.

The SAM IoT Wx v2 development board - EV75S95A uses SAM D21/DA1 family's ATSAMD21G18A MCU that is a Low-Power, 32-bit Cortex-M0+ MCU with Advanced Analog and PWM. It is generally used with google cloud. The WG development is designed for connecting to Google cloud.

References:

https://ww1.microchip.com/downloads/aemDocuments/documents/MCU32/ProductDocuments/Brochures/32-bit-Brochure-30009904.pdf

https://www.microchip.com/en-us/products/microcontrollers/32-bit-mcus/pic32-sam

2.6.6 Other Microchip series – Atmel, AVR & other Products

Some of the popular Atmel/AVR MCUs are the ATmega328P, ATmega32u4, ATmega2560 and Atmel SAM3X8E.

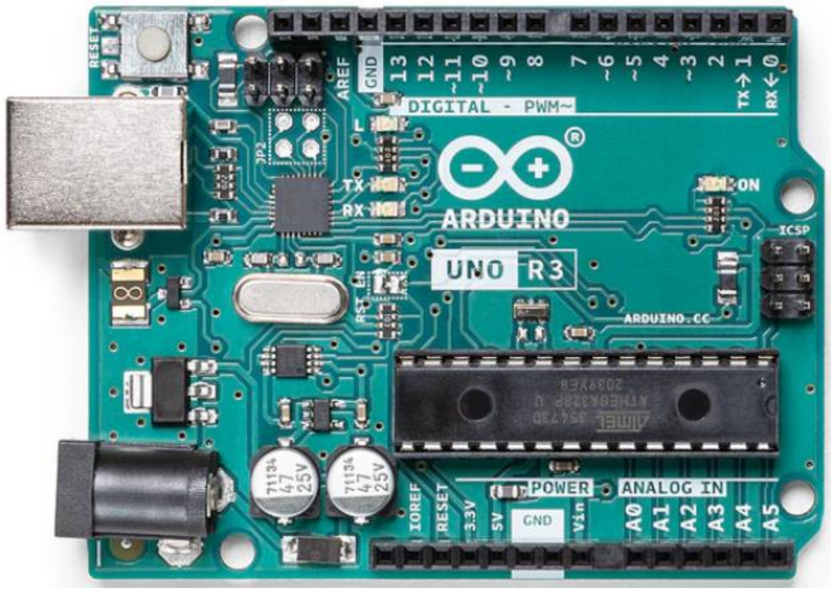

Fig 2.4: Arduino Uno R3 Board

Ref: https://store-usa.arduino.cc/collections/uno/products/arduino-uno-rev3

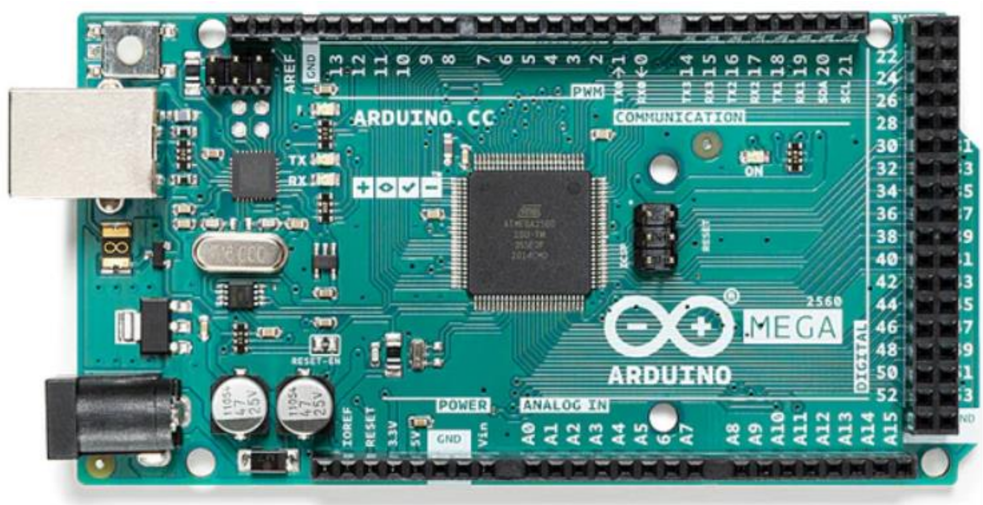

Fig 2.5: Arduino Mega2560 Board

Ref: https://store-usa.arduino.cc/collections/giga/products/arduino-mega-2560-rev3

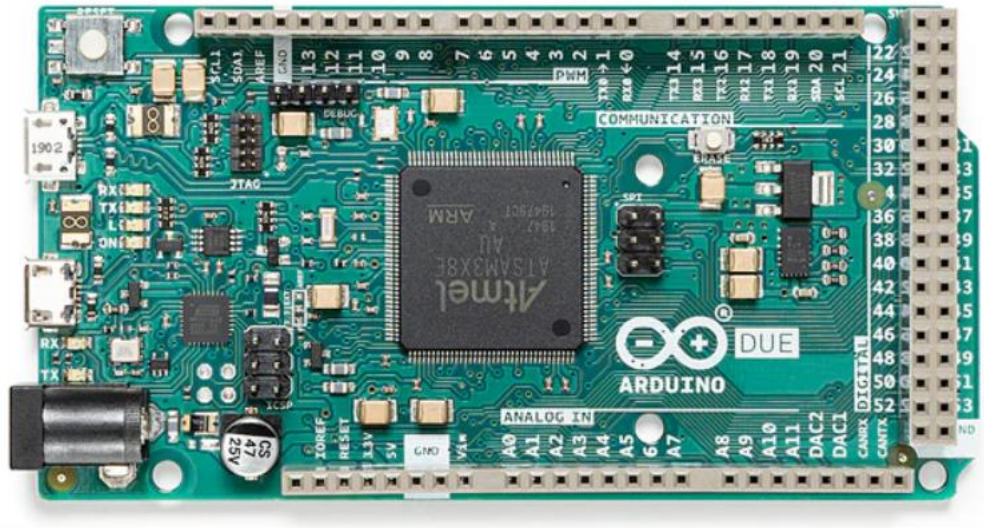

Fig 2.6: Arduino Due Board

Ref: https://store-usa.arduino.cc/collections/uno/products/arduino-due

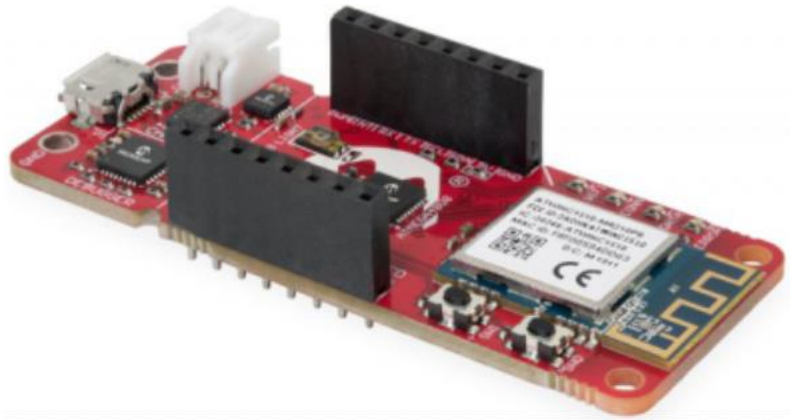

Fig 2.7: AVR IOT WA Board

Ref: https://www.microchip.com/en-us/development-tool/ev15r70a

One of the most popular open-source board the Arduino Uno uses the ATmega328P MCU. Similarly, other popular Arduino boards like Arduino Leonardo uses the ATmega32u4 MCU, the Arduino Mega 2560 uses the ATmega2560 MCU and Arduino Due uses the Atmel SAM3X8E MCU based on ARM Cortex-M3 CPU.

The AVR EA, EB, DA, DB, DD, DU, SD series are generally 8-bit devices. The mega series are 16-bit devices. The 32-bit AVR & RISC MCUS are based on AVR32 CPU, ARM7 CPU and SAM3 series based on Cortex-M3 CPU.

The AVR-IoT WA Development Board for Wi-Fi® Connection to AWS IoT Core - EV15R70A and AVR-IoT WG Development Board – AC164160 uses the ATmega4808 AVR MCU. These could be connected to AWS, Google clouds respectively.

Users are encouraged to refer the respective datasheets for the specifications and architectural features of the devices.

dsPIC Series

Microchip products also include digital signal controllers – DSC's or dsPIC's. The dsPIC's integrate the DSP functionality and the simplicity of MCU's. These are used in embedded DSP performance applications that require time-critical response and the simplicity of an MCU. These are dsPIC33A series that is 32-bit and dsPIC33C & dsPIC33E series that are 16-bit devices.

Ref: https://www.microchip.com/en-us/products/microcontrollers/dspic-dscs

Microchip has other series like PIC10, PIC12, PIC17 that vary between 12-bit to 16-bit wide code memory range with PIC10 and PIC12 being generally 12-bit wide code memory while PIC 17 being 16-bit wide. Apart from these Microchip products include several microprocessors up to 64-bit like PIC64 with multi core processors, FPGAS etc.